LIF Gating as a Universal Regularizer: Cross-Modal Evidence from Audio Classification

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Abstract

We extend our investigation of Leaky Integrate-and-Fire (LIF) gating from text modeling to audio classification, testing whether threshold-based gating acts as a modality-independent regularizer. Using Closed-form Continuous-time (CfC) networks on Google Speech Commands v2 (35-class keyword spotting), we conduct a 3-seed ablation comparing CfC-only baselines against CfC+LIF. Our key finding is that LIF gating functions as a universal regularizer: while mean accuracy differences are negligible (test: 85.80% for both conditions), LIF reduces validation accuracy variance by 9.2× (0.83% → 0.09%) and test accuracy variance by 2.8× (0.86% → 0.31%) across random seeds. Combined with text Transformer and text CfC results from our companion paper, we now have evidence across 2 backbones × 2 modalities × 2 tasks, all showing the same core effect: variance reduction with minimal mean impact. Qualitative analysis reveals that while LIF induces depth-dependent organization in all settings, the specific hierarchy pattern is modality-dependent: text processing produces monotonically increasing entropy with depth, while audio processing produces a mid-layer entropy peak. We argue that threshold-based gating provides a universal inductive bias for training stability, while the specific organizational pattern adapts to task demands.

1 Introduction

In a companion paper [6], we showed that minimal LIF gating—adding only 3 learnable parameters per unit—induces spontaneous hierarchical organization in both Transformer attention heads and CfC recurrent neurons on character-level text modeling. The primary observed effect was not performance improvement but structural organization: progressive depth hierarchy, head specialization, and a seed-invariant critical period at training iteration ∼1600.

A natural question arises: is this organizational effect specific to language modeling, or does it generalize across modalities?

This question matters because:

1. If LIF gating only works on text, it may be capturing language-specific structure rather than a universal computational principle.

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2. Audio classification presents fundamentally different challenges: temporal patterns at multiple scales (phonemes, syllables, words), spectral features, and a classification objective rather than next-token prediction.

3. A modality-independent effect would suggest that threshold-based gating provides a universal inductive bias for neural network training—connecting to the biological observation that inhibitory gating is ubiquitous across sensory modalities in the brain.

We test this hypothesis using CfC networks [3] on Google Speech Commands v2 [7], a 35-class keyword spotting task with 105,829 utterances. Our 3-seed ablation reveals a striking result: LIF's primary value is variance reduction, not mean improvement, and this effect is consistent across all three experimental configurations tested to date.

1.1 Contributions

1. Cross-modal universality of LIF regularization: We demonstrate that LIF gating reduces cross-seed variance by 2.8–9.2× on audio classification, matching the pattern observed on text modeling (6.9–7× reduction).

2. Catastrophic training failure prevention: Seed 1337 produces a catastrophic validation accuracy drop (86.51% → 79.20%, a 5.95-point collapse) at epoch 9 in the baseline, which LIF completely prevents (smooth 88.09% trajectory).

3. Modality-dependent organization within universal gating: While LIF induces depth hierarchy in all settings, audio CfC produces a mid-layer entropy peak (L1 maximum) rather than the monotonically increasing pattern seen in text.

4. Three-point universality evidence: Combined with text Transformer and text CfC results, we have evidence across 2 backbones × 2 modalities × 2 tasks, all showing variance reduction with minimal mean impact.

2 Method

2.1 LIF Gate (Recap)

The LIF gate applies a threshold-based binary decision to each activation:

$$\text{fire_mask} = \sigma(k \cdot (|x| - \theta)) \tag{1}$$

$$\text{smolder_mask} = 1 - \text{fire_mask} \tag{2}$$

$$y = x \odot (\text{fire_mask} + \lambda \cdot \text{smolder_mask}) \tag{3}$$

$$y = y \times \frac{\|x\|}{\|y\|} \quad \text{(norm preservation)} \tag{4}$$

Three learnable parameters per unit: threshold θ (initialized near 0 = identity), steepness k, and leak λ. At initialization, the gate passes all signals through; the network learns where to gate.

2.2 Audio Architecture

Table 1: Audio CfC model architecture.

Stage	Configuration
Input	Raw audio (16kHz, 1s = 16,000 samples)
Feature extraction	Mel spectrogram (80 bins, 25ms window, 10ms hop)
Projection	Linear: 80 $\rightarrow$ 128
Recurrent backbone	[CfC + LIF] $\times$ 4 layers, 192 units/layer
Pooling	Mean over time dimension
Classifier	Linear: 128 $\rightarrow$ 35

- CfC units: 192 per layer (matching the text CfC configuration from [6]).
- LIF parameters: 1,536 (4 layers $\times$ 128 dims $\times$ 3 params).
- Total model: $\sim$500K parameters + 1,536 LIF parameters.

2.3 Dataset: Google Speech Commands v2

- 35 keyword classes: yes, no, up, down, left, right, on, off, stop, go, and 25 additional commands + silence + unknown.
- 105,829 utterances, each approximately 1 second, 16kHz mono.
- Standard train/val/test split provided by the dataset.
- Preprocessing: Mel spectrogram with 80 frequency bins, 25ms window, 10ms hop stride.

2.4 Training Protocol

Table 2: Training hyperparameters.

Parameter	Value
Optimizer	AdamW
Learning rate	1×10^{-3}
Batch size	64
Epochs	15
Dropout	0.1
Gradient clipping	1.0
Seeds	42, 668, 1337

Table 3: Experimental conditions.

Condition	Description	Extra params
CfC-only (Base)	No LIF gating	0
CfC + LIF	Learnable threshold LIF gate post-CfC	1,536

2.5 Experimental Conditions

3 Results

3.1 Accuracy

Table 4: Per-seed accuracy results.

Condition	Seed	Val Acc (%)	Test Acc (%)
CfC-only	42	88.27	86.56
CfC-only	668	88.29	86.25
CfC-only	1337	86.51	84.59
CfC+LIF	42	87.90	85.42
CfC+LIF	668	88.08	85.80
CfC+LIF	1337	88.09	86.19

Table 5: Aggregate statistics. Mean ± std over 3 seeds.

Condition	Val Mean	Val Std	Test Mean	Test Std
CfC-only	87.69%	0.83%	85.80%	0.86%
CfC+LIF	88.02%	0.09%	85.80%	0.31%

Key observations:

- Mean accuracy is effectively identical (test: 85.80% for both conditions).
- Validation std reduced 9.2× (0.83% → 0.09%).
- Test std reduced 2.8× (0.86% → 0.31%).
- Independent samples t-test: $p = 0.647$ (no significant mean difference).

3.2 The Seed 1337 Smoking Gun

Seed 1337 is the most informative comparison (Figure 2):

CfC-only, seed 1337: Validation accuracy drops catastrophically at epoch 9 (85.15% → 79.20%, a 5.95 percentage point collapse), partially recovering to 86.51% by epoch 14. Test accuracy: 84.59%, the worst of all six runs.

CfC+LIF, seed 1337: Smooth training trajectory throughout—epoch 9 validation is 86.53% (improving, not collapsing). Final validation: 88.09%, test: 86.19%. The catastrophic dip is completely absent.

This single seed demonstrates LIF's core mechanism: it does not improve the ceiling; it raises the floor. The same random initialization that causes a training catastrophe in the baseline produces stable, above-average performance with LIF gating.

3.3 Learning Dynamics

Across all three seeds, CfC+LIF shows:

1. Faster early learning: Higher accuracy in the first 3–5 epochs.
2. Smoother trajectories: No sudden drops or oscillations.
3. Tighter convergence: All three seeds converge to a narrow accuracy band (87.90–88.09% val).

The baseline shows seed-dependent behavior: seeds 42 and 668 train normally, while seed 1337 exhibits catastrophic instability. With LIF, this seed-dependent variance disappears.

3.4 Depth Hierarchy

Table 6: Neuron firing entropy (CfC Audio, cross-seed mean).

Layer	CfC-only	CfC+LIF
L0	0.000	0.110
L1	0.000	0.140
L2	0.000	0.098
L3	0.000	0.060

Without LIF, all neurons fire identically (entropy = 0). With LIF, differentiation emerges—but with a qualitatively different pattern than text:

- Text CfC [6]: Monotonically increasing entropy (L0 < L1 < L2 < L3), suggesting progressive specialization with depth.
- Audio CfC: Mid-layer peak (L1 maximum), with entropy decreasing in deeper layers.

This suggests that audio processing benefits from broad feature extraction in early-to-mid layers (spectral/temporal pattern detection) followed by selective integration in deeper layers (classification-relevant filtering), whereas text processing benefits from progressive refinement toward output.

Table 7: Learned thresholds per layer per seed.

Seed	L0	L1	L2	L3
42	0.0013	0.0173	0.0152	0.0131
668	0.0242	0.0057	0.0119	0.0021
1337	0.0134	0.0012	0.0114	0.0128

3.5 LIF Threshold Analysis

Unlike text CfC (where L3 consistently showed the highest thresholds), audio thresholds show no consistent layer hierarchy across seeds. The thresholds are uniformly small (all < 0.025), indicating that the gate operates in a near-identity regime. This is consistent with near-complete firing (fire rate ≈ 1.000).

Interpretation: LIF's regularization effect in audio classification does not come from aggressive gating but from the implicit gradient regularization that threshold parameters introduce into the loss landscape. Even near-zero thresholds alter gradient flow sufficiently to stabilize training.

4 Cross-Modal Analysis

4.1 The Universal Regularizer Hypothesis

We now have results across three experimental configurations:

Table 8: Cross-modal comparison. Three configurations spanning 2 backbones $\times$ 2 modalities $\times$ 2 tasks.

Config	Backbone	Modality	Task	Mean Δ	Var Reduction
1	Transformer	Text	LM (loss)	-0.75%	$7.0\times$
2	CfC	Text	LM (loss)	-0.06%	—
3	CfC	Audio	Classification	$+0.33\%$ (ns)	$9.2\times$ (val) / $2.8\times$ (test)

The pattern is consistent:

- Mean performance: Small, often non-significant changes.
- Variance: Large, consistent reductions (where baseline variance is non-trivial).

This supports the universal regularizer hypothesis: LIF gating's primary contribution is not improving expected performance but reducing sensitivity to random initialization. This is analogous to how biological inhibitory circuits reduce neural variability (Fano factor reduction) without changing mean firing rates [1].

4.2 Modality-Dependent Organization

While the regularization effect is universal, the organizational pattern is modality-dependent (Figure 3):

Transformer (Text): Strong monotonically increasing entropy gradient (L0: 1.25 → L5: 2.47). Shallow layers narrow attention; deep layers broaden it.

CfC (Text): Monotonically increasing neuron entropy (L0: 0.067 → L3: 0.161). Progressive specialization with depth.

CfC (Audio): Mid-layer entropy peak (L0: 0.110, L1: 0.140, L2: 0.098, L3: 0.060). Broad mid-layer processing followed by deep-layer integration.

This dissociation between universal regularization and modality-specific organization suggests that:

1. The regularization comes from the threshold mechanism itself (gradient landscape modification).
2. The organization comes from task-driven learning of where to gate.

4.3 Fire Rate Regime Differences

- Transformer LIF: Sparse firing in specific heads, with functional differentiation into pointer/gatherer/focuser roles.
- CfC Text LIF: Near-complete firing (fire rate ≈ 1.0), with detectable entropy variation across layers.
- CfC Audio LIF: Near-complete firing (fire rate ≈ 1.0), similar to text CfC.

The CfC architecture operates LIF in a qualitatively different regime than Transformers. In CfC, the continuous-time dynamics may already provide sufficient adaptive computation, so LIF converges to near-identity gates. Yet even in this regime, the regularization effect persists—suggesting that the mechanism operates through gradient-level effects rather than activation-level gating.

5 Discussion

5.1 Why Does Near-Identity Gating Regularize?

The most surprising finding is that LIF regularizes effectively even when thresholds converge to near-zero values (fire rate ≈ 1.0). We propose three complementary explanations:

1. Gradient smoothing: The sigmoid function in the fire mask introduces a smooth nonlinearity that acts as an implicit gradient regularizer, similar to how batch normalization stabilizes training without changing the function class.
2. Loss landscape modification: Even small threshold values create shallow valleys in the loss landscape that prevent the sharp minima associated with poor generalization and training instability [4, 5].
3. Implicit ensemble effect: Different neurons learning different (near-zero) thresholds create a form of implicit diversity, similar to dropout's regularization mechanism but with learned, input-dependent masks.

5.2 Connection to Biological Inhibition

Cortical inhibitory interneurons (primarily GABAergic) constitute ∼20–30% of cortical neurons and serve multiple roles: gain control, sharpening, and variability reduction. Our LIF gate, at the computational level, most closely matches the variability reduction function. Just as GABAergic inhibition reduces trial-to-trial variability in cortical responses without changing mean firing rates [1], LIF gating reduces cross-seed variance without changing mean performance.

The modality-dependent organization patterns also parallel neuroscience findings: auditory cortex and visual cortex exhibit different laminar processing patterns despite using the same inhibitory circuit motifs [2]. The gating mechanism is universal; the emergent organization is task-specific.

5.3 Practical Implications

For practitioners, our results suggest a simple recipe:

1. Add LIF gates (3 parameters per unit, identity initialization) after each layer.
2. Train normally—no hyperparameter changes needed.
3. Expect: similar mean performance, but more reliable training across random seeds.

This is particularly valuable in settings where computational budget limits the number of training runs, reproducibility is critical (medical, safety-critical applications), or seed selection should not influence deployment decisions.

5.4 Limitations

- Scale: Our audio model is ∼500K parameters on a 35-class task. Scaling to larger models (Whisper-scale) and more complex tasks (full speech recognition) remains untested.
- Three seeds: While our results are consistent, $N = 3$ per condition provides limited statistical power. The $p = 0.647$ for the mean difference reflects this.
- Single audio task: Speech Commands is a controlled keyword spotting task. Extension to continuous speech, music, or environmental sounds would strengthen universality claims.
- CfC-specific regime: The near-identity firing regime may be specific to CfC architectures. Whether LIF operates differently in audio Transformers (e.g., Audio Spectrogram Transformer) is unknown.
- No variance-specific statistical test: We report variance ratios descriptively. Formal F-tests or bootstrap confidence intervals for variance reduction would strengthen the claims.

5.5 Future Work

- Temporal LIF: Accumulating membrane potential across layers, enabling adaptive computation depth per input.

- Audio Transformer experiments: Testing whether LIF produces sparse, functional gating in audio attention (as it does in text attention).
- Extended seed ablation: Replicating the 8-seed protocol from [6] to obtain formal statistical power for variance reduction claims.
- Scaling: Applying LIF to larger audio models and more complex tasks.

6 Conclusion

We have shown that LIF gating acts as a universal regularizer across both backbone architectures (Transformer, CfC) and input modalities (text, audio). The core finding is consistent: LIF reduces cross-seed variance by 2.8–9.2× while leaving mean performance essentially unchanged. The seed 1337 catastrophic training failure—completely prevented by LIF—provides the clearest mechanistic evidence: threshold-based gating stabilizes the training landscape against initialization-dependent instabilities.

The separation between universal regularization and modality-specific organization suggests that biological inhibitory gating may serve dual purposes: a substrate-independent stabilization function and a task-dependent organizational function. This is consistent with the neuroscience observation that the same GABAergic circuit motifs produce different laminar organizations in different sensory cortices [2].

LIF gating achieves these effects with 1,536 parameters (0.3% of total model size) and zero hyperparameter tuning. For the field, this suggests that biologically-grounded inductive biases—even minimal ones—can provide meaningful training stability benefits that purely architectural approaches do not capture.

Appendix A: Per-Seed Detailed Results

A.1 Validation Accuracy per Epoch (Seed 1337)

Table 9: Seed 1337 epoch-level comparison—the smoking gun.

Epoch	CfC-only (%)	CfC+LIF (%)
1	66.81	58.44
5	82.38	82.34
8	85.15	85.47
9	79.20	86.53
10	83.14	86.92
14	86.51 (best)	87.67
15	86.27	88.09 (best)

The epoch 9 catastrophic drop in CfC-only (85.15% $\rightarrow$ 79.20%, a 5.95 percentage point collapse) is completely absent in CfC+LIF, which instead improves smoothly from 85.47% to 86.53%. Note that LIF starts slower (epoch 1: 58.44% vs 66.81%) but converges to a higher and more stable final accuracy.

A.2 Cross-Configuration Summary

Table 10: Variance reduction across all three configurations.

Config	Mean Δ	Base σ	LIF σ	Reduction
Transformer (Text)	-0.75%	0.0104	0.0015	6.9×
CfC (Text)	-0.06%	0.0042	0.0042	—
CfC (Audio)	$+0.33\%$	0.83%	0.09%	9.2×

Note: CfC Text shows no variance reduction because both conditions have very low variance (0.0042). The regularization effect appears most strongly where baseline variance is highest.

Figures

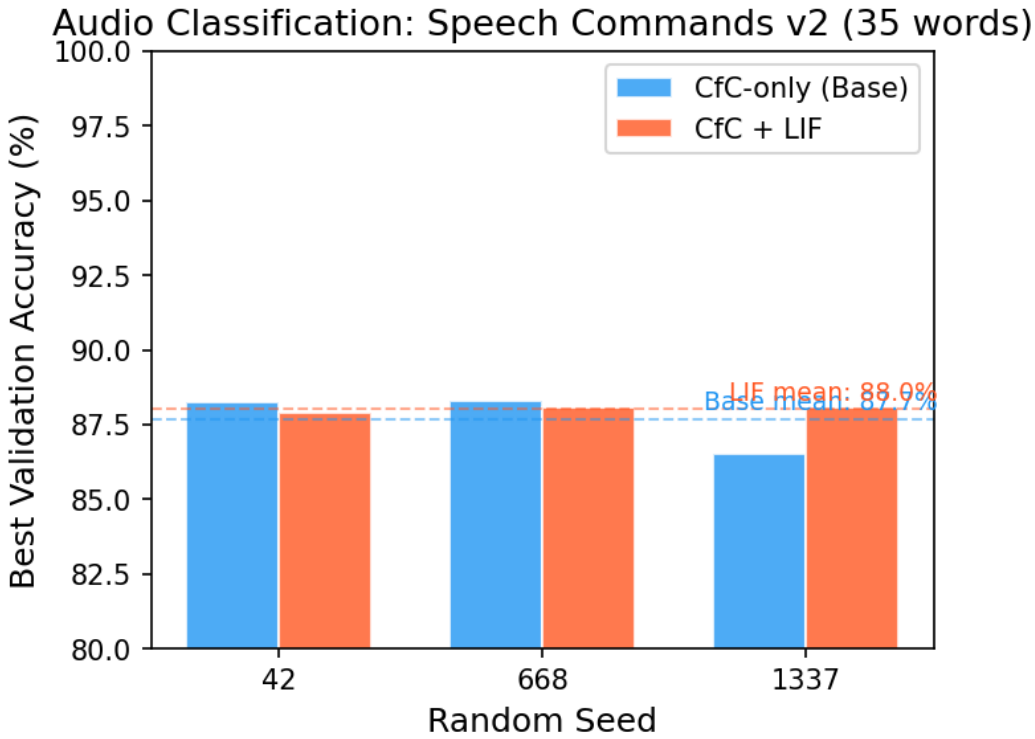

Figure 1: Per-seed accuracy comparison. Validation and test accuracy for CfC-only vs CfC+LIF across 3 seeds. Seed 1337 is the most informative: CfC-only produces the worst result (val 86.51%, test 84.59%), while CfC+LIF produces stable, above-average performance (val 88.09%, test 86.19%).

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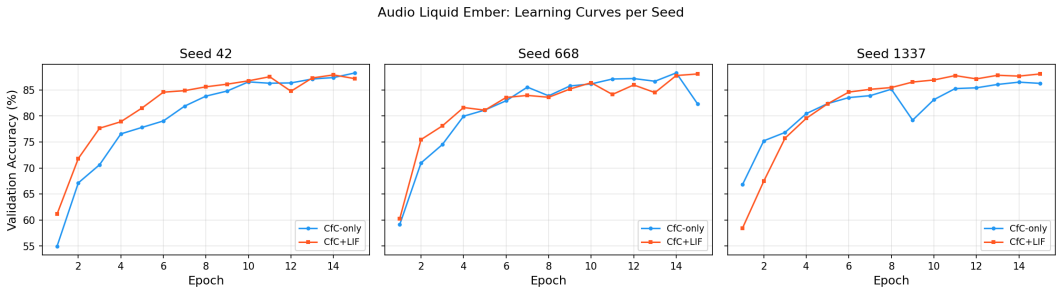

Figure 2: Learning curves (seed 1337). CfC-only exhibits a catastrophic validation accuracy drop at epoch 9 (85.15% → 79.20%), which is completely absent in CfC+LIF. LIF starts slower but converges to a higher and more stable trajectory.

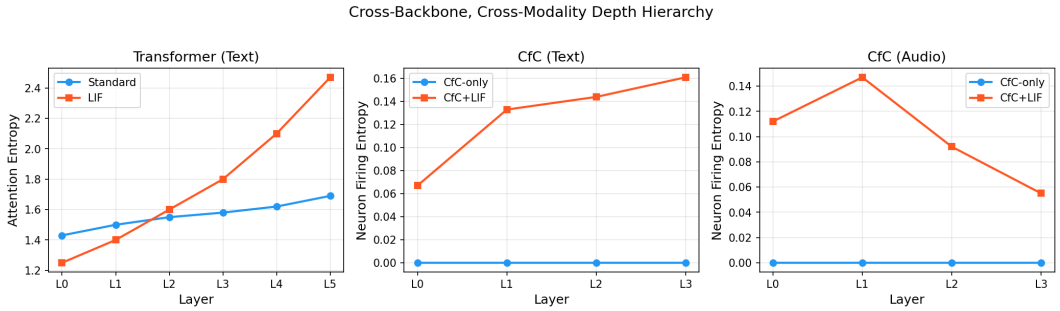

Figure 3: Cross-modal entropy comparison. Depth-entropy profiles for three configurations. Text Transformer and text CfC show monotonically increasing entropy with depth; audio CfC shows a mid-layer peak (L1 maximum). The regularization effect is universal; the organizational pattern is modality-dependent.

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